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FinFET Technology Based Development of Energy Efficient Datapath System

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***Abstract* - In today's electronics industry, extensive research is dedicated to advancing nanoscale VLSI designs, particularly focusing on energy-efficient solutions. Nanoscale designs often exhibit elevated power consumption due to the close proximity of source and drain regions within these diminutive devices. The limited channel space between source and drain weakens channel control, resulting in increased leakage currents. The FinFET device emerges as a prime candidate for contemporary electronic circuits, offering enhanced performance. This study is centered on the implementation of FinFET technology-based data paths, emphasizing low-power and high-speed capabilities. The data path assumes a crucial role in executing logical and arithmetic operations within the system, encompassing essential functions like NOT, AND, OR, and more. A 4-bit carry-look-ahead adder is employed for bitwise arithmetic tasks such as addition, subtraction, increment, and decrement. The designs are realized using the LT-Spice simulator, showcasing notable achievements. The proposed design significantly reduces power consumption, cutting the propagation delay by 9 µsec compared to CMOS technology, which consumed 4.7 µW. The findings strongly indicate that FinFET-based data paths outperform bulk CMOS technology, reinforcing their potential for superior performance and energy efficiency.**

Keywords—Datapath System, FinFET, LT Spice, Carry Look Ahead Adder.

I. INTRODUCTION

With the continuous progress of nanotechnology, the performance of CMOS technology has encountered limitations. This has resulted in diminished control over channels at the gate level, leading to substantial leakage currents even when devices are in an off-state. The outcome is an increase in static power consumption and undesirable heat generation within integrated circuits. In response to this challenge, research efforts have given rise to novel devices like double gate MOSFETs, Single Electron Transistors (SETs), and FinFETs, heralded as superior alternatives to CMOS devices in nanoscale designs [1].

Particularly, FinFETs have found utility in high-speed switching scenarios, garnering extensive research interest for their potential in areas like smart phones and supercomputers [2]. Their distinct characteristics, marked by elevated ON-current, speed, and minimal leakage, distinguish them from other options. Furthermore, the FinFET's qualities such as low dopant variation and enhanced carrier mobility enable scalability down to 32nm and beyond [3].

At the core of processor and computer systems, the Central Processing Unit (CPU) relies on robust data path architecture to facilitate parallel processing. This architecture, depicted in figure 1, performs essential arithmetic (addition, multiplication, subtraction, division) and logical operations (NOT, AND, OR, XOR). FinFET technology assumes a crucial role in the intricate design of datapath systems tailored to modern applications. The paper delves into the creation and presentation of power-efficient, high-speed complex datapath systems [4]. Through this exploration, the paper contributes to the ongoing advancement of sophisticated computing solutions, underscored by FinFET technology-based data path designs.

A 4-bit adder circuit designed with nanoscale technology is at the heart of the datapath system. This design not only showcases the fundamental addition operation (A+B), but also extends its capabilities to encompass subtraction and various logical functions. The (A-B) subtraction operation is achieved through the complementation of B followed by addition to A input. Additionally, the implementation of logical functions adheres to established equations, specifically from Eq. (1) to Eq. (3).

$$\text{Half adder sum } (S_H) = \overline{A_K} B_K + A_K \overline{B_K} \quad \text{--- (1)}$$

$$\text{Full adder sum } (S_F) = \overline{S_H} C_{K-1} + S_H \overline{C_{K-1}} \quad \text{--- (2)}$$

$$\text{Full adders carry } (C_F) = A_K B_K + S_H C_{K-1} \quad \text{--- (3)}$$

The fundamental structure of the datapath system is illustrated in figure 1. Within this system, the control unit incorporates blocks such as a multiplexer, AND gate, OR gate, and Full adder.

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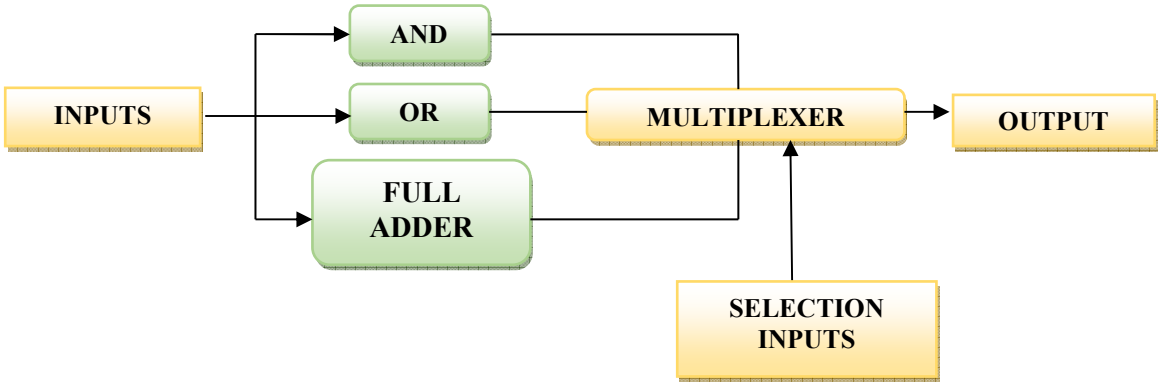


Fig. 1. Datapath's Primary Components

II. LITERATURE REVIEW

As asserted by G.Kumar et al., FinFETs offer superior performance in terms of voltage, area, and power when compared to CMOS technology [5]. The efficacy of current control relies on the fins situated on the channel of a FinFET device, regulating the current flow. CMOS gate control loss exacerbates challenges such as parameter variations, leakage current, reliability issues, short channel effects and yield concerns. Notably, SRAM cells commonly serve as provisional data storage before power supply activation [6], making the reduction of power consumption and enhancement of delay tolerance paramount objectives. The fusion of FinFET technology and adiabatic logic effectively meets these specifications [7].

The Charge Sharing Complementary Pass Transistor Adiabatic Logic (CSCPAL) with 4-phase power clock was presented by K. G. Jayashree et al. When compared to current FinFET based 2N2P, 2N2N2P and PFAL circuit equivalents, the design, which was simulated using 32nm FinFET models using Cadence Virtuoso design tools, performs better [8].

Utilizing 32nm MOSFETs and FinFETs, U.Tripathi et al constructed a comprehensive 28-transistor adder CMOS circuit [9]. Employing simulation in the HSPICE program, the delay and average power consumption of the adder were calculated for both FinFET and MOSFET configurations. The results showcased a reduction in both average power consumption and latency with the implementation of FinFET technology. The prospect of sustaining the process using DGFETs was also explored. Remarkably, the scaling of DGFET technology is led by the advancements in FinFET technology, given their straightforward manufacturability and broad utility.

In summary, FinFETs demonstrate a clear advantage over CMOS in terms of voltage, area, and power performance, attributed to their effective current control through channel fins. Research has yielded innovative designs like the CSCPAL circuit, showcasing superior performance compared to existing counterparts. Moreover, investigations into comprehensive adder circuits using FinFET and MOSFET technologies underscored the substantial benefits of FinFETs, with potential extensions into DGFET technology driven by their manufacturing simplicity and applicability.

III. Fin FIELD-EFFECT TRANSISTOR (FinFET) TECHNOLOGY

FinFET technology introduces a distinctive perpendicular fin-shaped channel, departure from the parallel channel configuration of the planar MOSFET [10], depicted in figure 2.

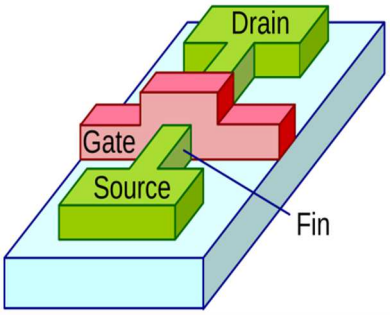


Fig. 2. FinFET Structure

A. Based on Number of Gates

A member of the FinFET family, the Double Gate FET (DG-FET) incorporates two gates relative to the channel. Figure 3 illustrates a cross-sectional view of the planar DG-FET. Although current is transmitted horizontally across the wafer, the device's shape in the vertical direction [11-12] (fin altitude) affects its behavior, earning it the moniker of a "quasi-planar" structure. By having front and back gate electrodes that are similar in the Symmetric DGFET [13], an inversion channel may be created around the borders of the plane. It is important to note that both the top and bottom gate conductors are built from polysilicon, but one is n+ and the other is p+.

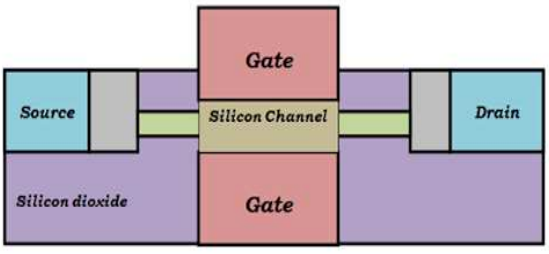


Fig. 3. DGFET Structure

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The dual-gate FinFET configuration is depicted in figure 4. The nomenclature stems from its resemblance to a fish fin due to the perpendicular and compact channel architecture. Lfin refers length of the gate, which is analogous to the typical FET channel length; Hfin refers height of the fin; and Wfin refers fin's thickness.

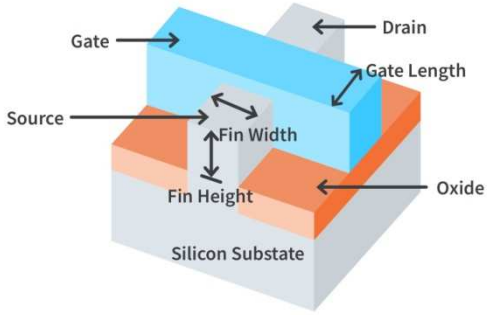


Fig. 4. Plan view of a FinFET transistor

B. *Wafer-based implementation*

Both the bulk FinFET structure and the Silicon on insulator (SOI) FinFET structures are shown in figure 5. Fins in SOI FinFET are completely segregated from physical touch, but in bulk FinFET they are incorporated into the silicon substrate (bulk).



Fig. 5. (a) Bulk FinFET (b) SOI FinFET

C. *Utilizing logic gates for connections*

Categorizing by gate manipulation, FinFET branches into Independent Gate FinFET (IGF) and Short Gate FinFET (SGF). Figure 6(a) and (b) depict the symbols for SGF and IGF, respectively. IGF is a four-terminal configuration where each gate is autonomously controlled. SGF, on the other hand, is a three-terminal setup with two gates interconnected.

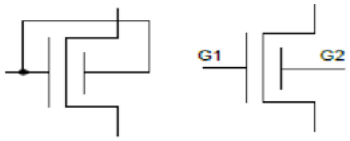


Fig. 6. (a) SGF symbol (b) IGF symbol

D. *Models of fin FETs*

Equations (4) to (6) embody the underlying physics within the FinFET drift-diffusion model utilized for simulations.

$$\nabla(-\in \nabla\varphi) = \rho \qquad \text{--- (4)}$$

Where $\rho = q[p - n + N_d - N_a]$

Current equations: $J_n = qn\mu_n \nabla\varphi + qD_n \nabla_n$ --- (5)

$$J_p = qn\mu_p \nabla\varphi + qD_p \nabla_p \qquad \text{--- (6)}$$

$$(\partial n)/(\partial t) = 1/q\nabla.J_n + R_n \qquad \text{--- (7)}$$

$$(\partial p)/(\partial t) = 1/q\nabla.J_p + R_p \qquad \text{--- (8)}$$

a. *Micro model*

Building upon the typical gate setup in multi-gate FETs, the development of BSIMCMG and BSIM-IMG models has facilitated the exploration of independent or asymmetric multi-gate FET configurations. In the case of DG-FinFETs, the physics-centric core models are informed by parameters like doping concentration and energy bands, grounded in physical principles.

Physically, FinFET is typically represented by metal gates with undoped fins and mid-gap function. Instead of using unused metal gates, the flat band voltages (V_{FB}) is used to define the threshold voltages (V_{TH}). The attention now goes to poly gates and doped fins. The forward and reverse gate voltages of a totally depleted thin body are calculated as follows, taking into consideration self-induced surface states.

$$V_{GFS} = V_{fb}\left(1+\frac{C_b}{C_{of}}\right)\phi_{sf} + \left(1+\frac{C_b}{C_{of}}\right)\phi_{sb} - \frac{Q_{ef}}{C_{of}} - \frac{Q_b}{2C_{of}} \qquad \text{--- (9)}$$

$$V_{GBS} = V_{FBb}\left(\frac{C_b}{C_{of}}\right)\phi_{sf} + \left(1+\frac{C_b}{C_{ob}}\right)\phi_{sb} - \frac{Q_{cb}}{C_{ob}} - \frac{Q_b}{2C_{ol}} \qquad \text{--- (10)}$$

Front and rear gate-to-source voltages are represented by VGFS and VGBS, respectively, and their expressions are included in Equations (9) and (10). These equations involve Q_{cf} and Q_b, which stand for the forward-facing and back-surface invert charge densities, a `1s well as sf and sb, which stand for the front and back surface potentials.

$$Q_b = -qN_A t_{Si} \qquad \text{--- (11)}$$

$$C_{of} = \frac{\in_{ox}}{t_{oxf}} \text{ and } C_{ob} = \frac{\in_{ox}}{t_{oxb}};\ C_b = \frac{\in_{si}}{t_{si}} \qquad \text{--- (12)}$$

For symmetrically driven DGFETs,

set $V_{GfS} = V_{GS}$ V_{GbS}.

Taking off the sb word yields to follow formula for V_{GS}:

$$V_{GS} = \phi_{sf} + \frac{1}{1+r}\left[\left(V_{FBf} + \gamma V_{FBb}\right)\right] - \left(\frac{Q_{ef}}{C_{of}} + \gamma\frac{Q_{cb}}{C_{ob}}\right) - \left(\frac{Q_b}{2C_{of}} + \gamma\frac{Q_b}{2C_{ob}}\right) \qquad \text{--- (13)}$$

Considering the thickness of the inversion channel and its associated with a capacitance, a threshold voltage (V_{TH}) of a slender, fully depleted asymmetrical DGFET is articulated through Equation (14).

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$$V_{TS} = \phi_{sf} + \frac{1}{1+r}\left[\left(V_{FBf} + \gamma W_{FBb}\right)\right] - \left(\frac{Q_b}{C_{of}} + \gamma \frac{Q_{cb}}{C_{ob}}\right) \quad \text{--- (14)}$$

IV. DESIGN CONCEPT UTILIZING FINFET TECHNOLOGY

Proposing a datapath system architecture rooted in 32nm FinFET technology, this design facilitates essential logical operations (AND, OR, XOR, NOT). Accompanying this, a 4-bit adder conducts fundamental arithmetic tasks, encompassing bitwise addition and subtraction. At the core of this architecture lie logic gates, the fundamental components of the datapath system. Illustrated in figure 7(a) to 7(e) are 32nm FinFET technology-based implementations of key logic gates: inverter, OR, AND, and Multiplexer.

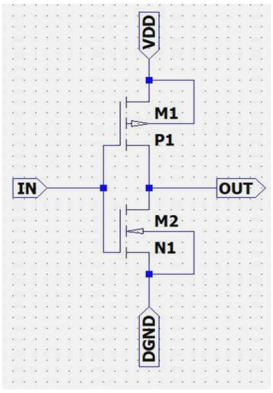


Fig. 7. (a) FinFET Technology based Implementation of Inverter

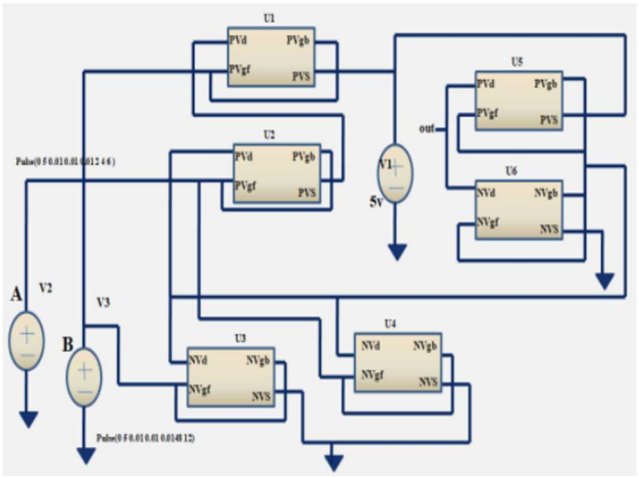


Fig.7. (b) Applying an OR gate in a proposed circuit

The datapath system has two control inputs to determine which procedures should be run. In cases where the control signals stand at S0 = 0 and S1 = 0, the system undertakes logical OR operations. Similarly, when S0 = 0 and S1 = 1, a logical NOT operation is executed. Transitioning to S0 = 1 and S1 = 0, the system performs logical AND operations. Lastly, with S0 = 1 and S1 = 1, the operations pivot based on the entirety of the input Cin. A comprehensive truth table outlining these operations is showcased in table 1.

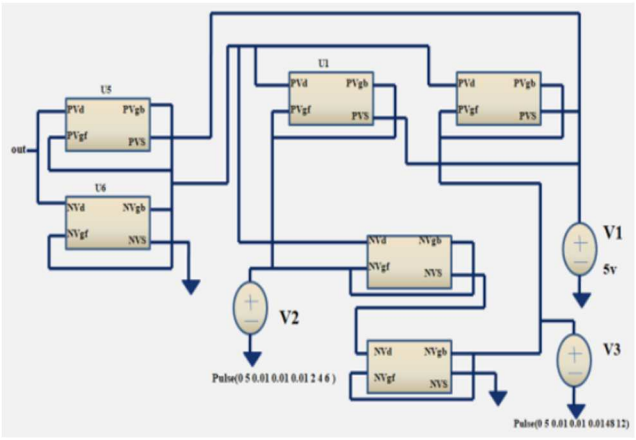


Fig. 7. (c) Developed with an AND gate

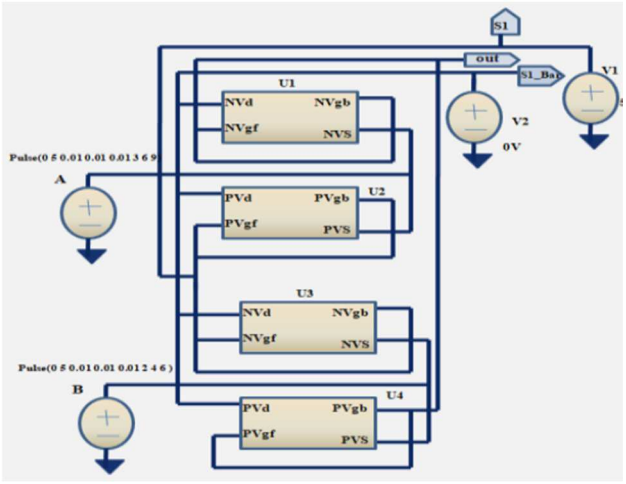


Fig. 7. (d) Incorporation of a 2 to 1 multiplexer

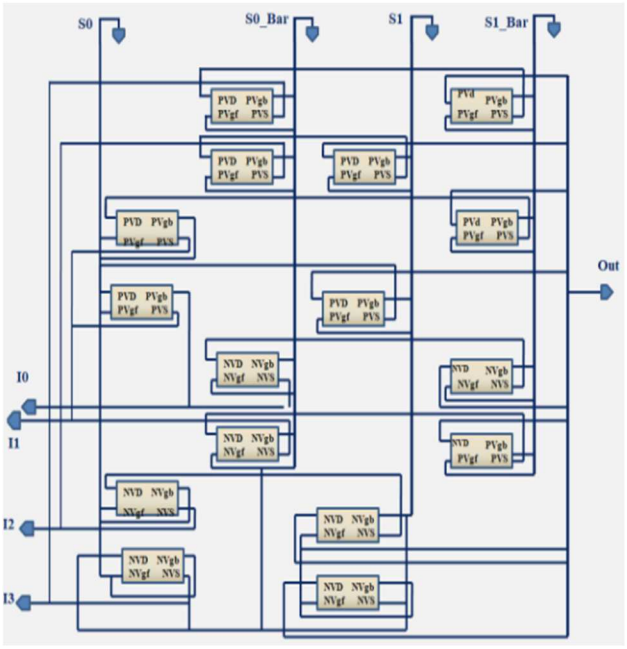


Fig.7. (e) Incorporation of the 4 to 1 Multiplexer

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Table 1. On the basis of selection lines, mathematical and logical operations processes were conducted

S0	S1	Circuit Function	Operation
1	1	Full Adder Sum	Arithmetic operations perform depends upon carry input C_{in}
1	0	AND	Logical
0	1	NOT	Logical
0	0	OR	Logical

V. RESULTS AND DISCUSSION

The data path system, meticulously crafted within the realm of 32nm FinFET technology, is visually encapsulated in figure 8. The resultant output waveform of this innovative data path system is visually presented in figure 9. This system finds its foundation in fundamental building blocks, namely the AND, OR gates, 4-bit adder, and 4-to-1 multiplexer, collectively employed to realize both arithmetic and logical functionalities. Notably, the data path system operates under a low-voltage regime, underscoring its advantageous performance relative to conventional CMOS technology, as elucidated in figure 10 and figure 11.

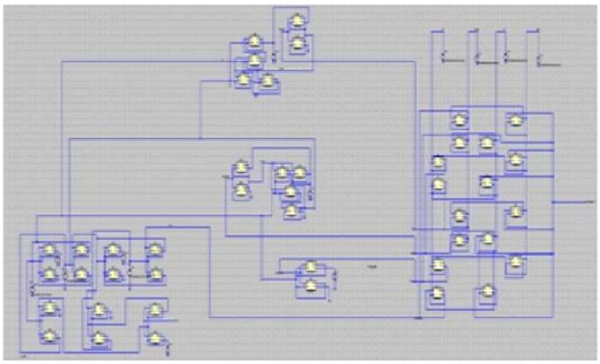


Fig. 8. Proposed Structure with FinFET Data Path

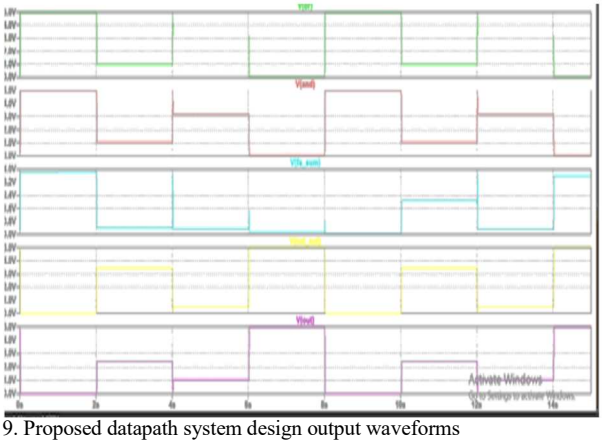


Fig. 9. Proposed datapath system design output waveforms

Provided in table 2 are comparative findings between the 32nm FinFET datapath system and the CMOS datapath counterpart. Impressively, the FinFET-based system exhibits a notable 63.28% decrease in average power dissipation, signaling its enhanced energy efficiency. The propagation latency within the FinFET datapath system registers a remarkable 95% reduction in comparison to the conventional CMOS architecture. The power delay product (PDP), it has been significantly reduced to a value of 0.004 pJ, is an indication of the continued progress that has been made in this area. Significant advancements may be seen in the Energy Delay Product (EDP), which is produced from the product of PDP and delay. The EDP showcases a substantial reduction when juxtaposed with standard CMOS technology, reinforcing the prowess of the FinFET-based design.

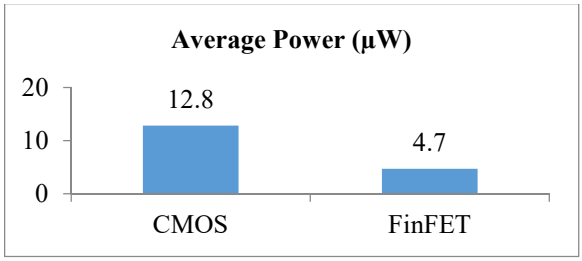


Fig. 10. Datapath's average power comparison of CMOS and FinFET

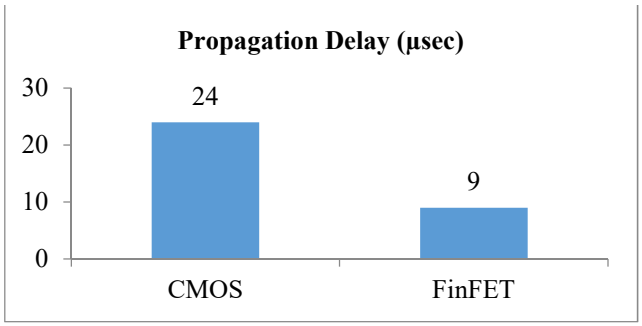


Fig. 11. Datapath's delay comparison of CMOS and FinFET

Table 2. Performance comparison of CMOS and FinFET datapath system circuits

Parameters	Existing (CMOS)	Proposed (FinFET)
Propagation Delay (µsec)	24	9
Average Power (µW)	12.8	4.7
Energy Delay Product (EDP) (Js)	$0.0073*10^{-18}$	$5.2*10^{-22}$
Power Delay Product (PDP) (PJ)	0.308	0.004

VI. CONCLUSION

Executed within the 32nm FinFET technology, the datapath system's core operations are comprehensively addressed within this study. This technology exhibits superior energy efficiency compared to CMOS counterparts, evident in the reduced average power consumption of 4.7µW as opposed to CMOS. The design process is carried out through the LTSPICE software, enabling effective evaluation of low on-chip power density strategies. Enhancing propagation time is achievable by leveraging the power source as a VDD source, promising tangible benefits. Furthermore, it is plausible to enhance the recommended design by assessing bit error rate and energy dynamics across diverse nodes. Such attributes position FinFET-based designs favorably for applications in telecommunications and health monitoring. Moreover, the developed Arithmetic Logic Unit (ALU) could serve as a foundational element, fostering the creation of higher-bit ALUs and sophisticated processing circuits, accentuating its utility in advancing computational capacities.

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